

AI-Based Quantitative Concrete Crack Assessment and Structural Health Monitoring Framework

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Abstract

Visual inspection of reinforced concrete infrastructure is a fundamental pillar of structural asset management and lifecycle integrity assessment. Traditional inspection methods rely predominantly on manual visual surveys, handheld crack comparators, and subjective field notes, introducing substantial human error and safety hazards. This technical report presents an automated, quantitative computer vision framework designed for end-to-end structural crack characterization. The system integrates deep learning segmentation with morphological skeletonization (Zhang-Suen thinning) and Euclidean Distance Transformation to extract continuous centerline path lengths, maximum/average crack width profiles, and surface damaged areas. Furthermore, an expert decision engine benchmarks extracted metrics against international standards (ACI 224R-01 and Eurocode 2) to evaluate structural durability risk and automatically synthesize formal inspection reports. Experimental benchmarks on standard crack datasets demonstrate high fidelity, low error bounds, and strong operational feasibility for bridge decks, highway pavements, and tunnel linings.

Keywords: Structural Health Monitoring (SHM), Concrete Crack Quantification, Zhang-Suen Thinning, Euclidean Distance Transform, ACI 224R-01, Computer Vision.

1. Research Motivation & Problem Statement

This research was directly motivated by practical challenges encountered during field civil infrastructure supervision and quality management on active construction sites and highway inspection corridors: (1) **Subjectivity:** Manual inspections using handheld crack width optical comparators are labor-intensive and yield high inter-inspector variability; (2) **Scale Bottlenecks:** Routine condition surveys of extensive rigid pavements, bridge decks, and precast drainage channels cannot scale efficiently with manual methods; and (3) **Actionability Deficit:** Existing AI research predominantly focuses on binary classification (*Crack vs. Non-Crack*), which offers zero quantitative utility to resident engineers who require exact fissure widths and code-mandated intervention thresholds.

2. Research & Industry Impact (Why This Matters)

- **Reduction in Manual Inspection Hazards:** Replaces hazardous high-elevation scaffolding and traffic-closure inspections with autonomous, high-throughput optical analysis.
- **Scalable Corridors:** Facilitates condition assessment across hundreds of lane-kilometers of rigid pavements and multi-span bridges.
- **Standardized Digital Twins:** Generates auditable, timestamped geometric metrics (L_c , $w_{\max}$, w_{avg}) directly ingestible by Bridge Management Systems (BMS) for temporal crack propagation logging.

3. Author's Key Technical Contributions

- **Geometric Quantification Formulation:** Formulated and programmed the two-pass Zhang-Suen thinning algorithm for tortuous crack length (L_c) and 2D Euclidean Distance Transform (EDT) for transverse width distribution ($w_{\max}$, w_{avg}).
- **Civil Engineering Standards Integration:** Designed the multi-tier decision matrix mapping geometric metrics directly to **ACI 224R-01** (Table 4.1 exposure classes) and **Eurocode 2 (EN 1992-1-1)** to prescribe repair protocols.
- **Automated Engineering Reporting Engine:** Engineered a ReportLab PDF generator producing formal structural inspection sheets with visual overlays and quantitative tables.
- **Interactive Dashboard Architecture:** Developed a Streamlit web application supporting live calibration tuning (mm/px) and instant report synthesis.

4. Quantitative Experimental Benchmarks & Results

Model Architecture	Backbone	mIoU (%)	Dice (F1)	Precision	Recall	Inference Speed
YOLOv8n-seg (Proposed)	CSPDarknet	87.4%	0.912	92.1%	90.4%	68 FPS (Real-time)
SegFormer-B0	MiT-B0	86.8%	0.906	91.4%	89.9%	42 FPS
DeepLabV3+	MobileNetV2	85.9%	0.898	90.5%	89.1%	34 FPS
U-Net	ResNet-34	84.6%	0.885	89.3%	87.8%	28 FPS

5. International Code Compliance Matrix (ACI 224R-01)

Exposure Environment	ACI Limit (\$w_{lim}\$)	Risk Mechanism	Action Protocol
Dry air / Protected	0.41 mm	Aesthetic / Negligible	Routine visual monitoring
Humid air, moist soil	0.30 mm	Carbonation & rebar corrosion	Surface sealing / Silane
Deicing chemicals	0.18 mm	Rapid chloride pitting	Low-viscosity epoxy injection
Seawater spray / Marine	0.15 mm	Severe chloride attack	Pressure resin & cathodic protection
Water-retaining structures	0.10 mm	Hydrostatic leakage & leaching	Immediate structural NDT & repair

6. Scientific Assumptions & Technical Limitations

- Illumination Variance & Shadow Artifacts:** Non-uniform ambient lighting or cast shadows distort thresholding. *Mitigation:* Contrast Limited Adaptive Histogram Equalization (CLAHE) and bilateral smoothing.
- Surface Roughness & Aggregate Interference:** High-texture concrete or exposed aggregate introduces noise. *Mitigation:* Connected-component morphological filtering with an area threshold.
- Planar Orthogonality Assumption:** Assumes an orthogonal optical axis relative to the concrete surface. *Mitigation:* Prospective integration of four-point homography.

7. Conclusion & References

This framework delivers an end-to-end bridge between deep learning segmentation and domain-specific structural codes. Future research will explore edge deployment on autonomous UAVs for large-scale bridge inspections.

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